

MTH401

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PAID SOVED CURRENT PAPER

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Differential Equations (MT11401)

Question: 2 (Marks: 1)

A point x_0 of a differential equation that is not an ordinary point is said to be

Choices:

☐ Non-singular point

☐ Singular point

☐ Limit point

☐ None of them

Differential Equations (MTH401)

Question: 1 (Marks: 1)

By using ratio test, $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ on any power series, the series is said to be neither convergent nor divergent if

Choices:

☒ $L < 1$

☐ $L > 1$

☐ $L = 1$

☐ None of them

Differential Equations (MTH401)

Question: 4 (Marks: 1)

Any linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 x \frac{dy}{dx} + a_0 y = g(x) \text{ where } a_0, a_1, a_2, \dots, a_n \text{ are constants}$$

is called

Choices:

☒ Homogeneous equation

☐ Polar equation

☐ Equi-dimensionl equation

☐ None of them

Differential Equations (MTH401)

Question: 3 (Marks: 1)

Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systematic elimination of the

Choices:

☐ Dependent variables

☐ Independent variable

☐ Choice variable

☐ None of them

Differential Equations (MTH401)

Question: 5 (Marks: 1)

The solution of the linear first order differential equation $\frac{dy}{dx} - 2y = 0$ is

Choices:

(a) $y = e^{2x+c}$



(b) $y = \sum_{n=0}^{\infty} \frac{x^n}{4n!}$



Both (a) and (b)



None of them



Differential Equations (MTH401)

Question: 6 (Marks: 1)

Ordinary points of $(x^2 - 64)(x^2 - 36)y'' + xy' - y = 0$ are

Choices:

☒ 0,1

☐

☐ 8,-8

☐

☐ 6,-6

☐

☐ None of others.

☐

Differential Equations (MTH401)

Question: 7 (Marks: 1)

The system of ordinary differential equations may involve two or more differential equations that contain derivatives of two or more dependent variables with respect to independent variable(s).

Choices:

one



two



three



arbitrary number of



Differential Equations (MTH401)

Question: 9 (Marks: 1)

If the equation of free damped motion have real and distinct roots, then system is _____.

Choices:

Over-damped

☐

Critically-damped

☐

Under-damped

☐

Weakly-damped

☐

Differential Equations (MTH401)

Question: 8 (Marks: 1)

The vectors $x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$ and $x_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$ are

Choices:

☒ linearly dependent

☐ linearly independent

Differential Equations (MTH401)

Question: 10 (Marks: 1)

The reactance and impedance both are measured in

Choices:

Newton

☐

Ohms

☐

Volts

☐

None of them

☐

TIME LEFT

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Differential Equations (MTH401)

Question: 11 (Marks: 1)

$\det(A - \lambda I) = 0$; this equation is called _____

Choices:

☒ eigen value

☐ characteristic equation

☐ eigen vector

☐ polynomial equation

Differential Equations (MTH401)

Question: 14 (Marks: 1)

The non-homogeneous system of differential equations

$$\frac{dx}{dt} = 5x - 7y + 5 \sin t$$

$$\frac{dy}{dt} = 7x + 5y - 11 \sin t$$

in the matrix form

Choices:

☐
$$\mathbf{x}' = \begin{pmatrix} 5 & -7 \\ -7 & -5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ 11 \end{pmatrix} \sin t$$

☐
$$\mathbf{x}' = \begin{pmatrix} 5 & -7 \\ 7 & -5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ 11 \end{pmatrix} \sin t$$

☐
$$\mathbf{x}' = \begin{pmatrix} 5 & -7 \\ 7 & -5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ -11 \end{pmatrix} \sin t$$

☐
$$\mathbf{x}' = \begin{pmatrix} 5 & -7 \\ 7 & 5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ -11 \end{pmatrix} \sin t$$

Differential Equations (MTH401)

Question: 13 (Marks: 1)

For finding the general solution of the non-homogeneous system of linear differential equations we need to find _____

Choices:

☐ complementary solution

☐ particular solution

☐ singular solution

☐ Both complementary and particular solution

Differential Equations (MTH401)

Question: 15 (Marks: 1)

Let t denote any point in some interval denoted by I . Then the problem of solving the system of differential equations

$$\frac{dX}{dt} = A(t)X + F(t)$$

Subject to the _____ conditions
 $X(t_0) = X_0$

Choices:

☐ dirichlet

☐ neumann

☐ initial

☐ boundary

Differential Equations (MTH401)

Question: 16 (Marks: 1)

$$\Gamma(n+1) =$$

Note: Γ is the gamma function

Choices:

☐ $(n-1)!$

☐ $n!$

☐ None of them

☐ $(n+1)!$

Differential Equations (MTH401)

Question: 17 (Marks: 1)

$x = 0$ is a(an) ~~regular~~ point of the differential equation: $y'' + xy' + 2y = 0$.

Choices:

☒ Ordinary

☐ Regular singular

☐ Singular

☐ Irregular singular

Question: 19 (Marks: 1)

If wroskian of the solution vectors X_1 and X_2 is zero, then vectors are -----.

Choices:

Parallel

☐

Linearly dependent

☐

Linearly Independent

☐

None of them

☐

Differential Equations (MTH401)

Question: 18 (Marks: 1)

The matrix $\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ is

Choices:

Scalar matrix

☐

Singular matrix

☐

Non-singular matrix

☐

Diagonal matrix

☐

Question: 22 (Marks: 1)

The eigenvalues of homogeneous system $X' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} X$, are

Choices:

☒ $\lambda_1 = 1, \lambda_2 = -1$

☐ $\lambda_1 = 1, \lambda_2 = -1$

☐ $\lambda_1 = 1, \lambda_2 = -1$

☐ $\lambda_1 = -1, \lambda_2 = -1$

Question: 20 (Marks: 1)

If $\begin{vmatrix} 2-\lambda & -8 \\ 2 & -6-\lambda \end{vmatrix} = 0$ then it gives

Choices:

$\lambda = -6$ of multiplicity 2

☐

$\lambda = 2$ of multiplicity 2

☐

$\lambda = 6$ of multiplicity 2

☐

$\lambda = -2$ of multiplicity 2

☐

Differential Equations (MTH401)

Question: 23 (Marks: 1)

The matrix $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ has

Choices:

☒ Complex eigen value

☐ Repeated & real eigen value

☐ None of them

☐ Real and unequal value

Question: 24 (Marks: 1)

If a matrix $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then eigenvalues are

Choices:

2,-1

☐

-1,-2

☐

1,-2

☐

2,1

☐

TIME LEFT

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Question: 21 (Marks: 1)

The matrix $\begin{bmatrix} \sqrt{3} & 1 & 2 \\ 0 & \sqrt{3} & 1 \\ 0 & 0 & \sqrt{3} \end{bmatrix}$ has eigenvalue of multiplicity

Choices:

1

☐

2

☐

3

☐

$\sqrt{3}$

☐

TIME LEFT

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Differential Equations (MTH401)

Question: 25 (Marks: 1)

If L denote the linear differential operators with constant coefficients, then $L_1 L_4 - L_2 L_3$ represents the

Choices:

☒ $\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$

☐ $\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$

☐ $\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$

☐ None of these

Question: 28 (Marks: 1)

What is the linear normal form of $y'' - y' + 6y = 0$, by using

$$\begin{aligned}y &= x_1, \\y' &= x_1' = x_2, \\y'' &= x_2'?\end{aligned}$$

Choices:

☒ $x_1' = x_2, x_2' = x_2 - 6x_1$

☐ $x_1 = x_2, x_2' = x_2 + x_1$

☐ $x_2' = x_2$

☐ $x_2' = x_2, x_2' = x_3 - 6$

Differential Equations (MTH401)

Question: 27 (Marks: 1)

What is the differential operator form of the differential equation $|a|y'' + by' + 2y = \cos x$?

Choices:

☒ $(bD + 2)y = \cos x$

☐ $(-aD^2 + bD + 2)y = \cos x$

☐ $(|a|D^2 + bD + 2)y = \cos x$

☐ $(aD^2 + bD + 2)y = \cos x$

Differential Equations (MTH401)

Question: 26 (Marks: 1)

If determinant $\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} = 0$, then which of the following is true?

Choices:

☒ $D^2 - 4D + 1 = 0$



$D^2 - 4D + 4 = 0$



$D^2 - 4D + 3 = 0$



None of these



Question: 29 (Marks: 1)

What are the eigen values for the matrix $\begin{pmatrix} 1 & 5 \\ 0 & 7 \end{pmatrix}$?

Choices:

☒ {1,5}

☐ {1,0}

☐ {6,5}

☐ {1,7}

Question: 32 (Marks: 1)

For the eigen value $\lambda = 5,5$ of the matrix $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$,
there exists _____ eigen vectors.

Choices:

☒ infinite

☐ one

☐ two

☐ three

Differential Equations (MTH401)

Question: 31 (Marks: 1)

What is the following differential equation called?

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

Choices:

☒ Cauchy's equation

☐ Newton's equation

☐ Legendre's equation

☐ Bessel's equation

Question: 30 (Marks: 1)

What are the eigen values for the matrix $\begin{pmatrix} 0 & 1 \\ 6 & 7 \end{pmatrix}$?

Choices:

☒ {0,7}

☐ {1,6}

☐ {-1,-6}

☐ {7,5}

Question: 33 (Marks: 1)

A singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ said to be a regular singular point if _____.

Choices:

☒ $(x - x_0)Q(x)$ is analytic at $x = x_0$

☐ $(x - x_0)P(x)$ is analytic at $x = x_0$

☐ $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at $x = x_0$

☐ $(x - x_0)^2P(x)$ and $(x - x_0)Q(x)$ are analytic at $x = x_0$

Question: 35 (Marks: 1)

$$J_{-\frac{2}{3}}(x) - J_{\frac{4}{3}}(x) = \underline{\hspace{2cm}}$$

Choices:

☐ $2J'_{\frac{4}{3}}(x)$

☒ $2J'_{\frac{2}{3}}(x)$

☐ $2J'_{\frac{8}{3}}(x)$

☐ $2J'_{\frac{1}{3}}(x)$

Question: 34 (Marks: 1)

By converting the equation $(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$ into the general form $y'' + P(x)y' - Q(x)y = 0$: $P(x)$ is _____.

Choices:

☒ $\frac{1}{(x-2)^2(x+2)^2}$

☐ $\frac{1}{(x+2)^2(x+2)^2}$

☐ $\frac{1}{(x-2)(x+2)}$

☐ $\frac{1}{(x-2)(x+2)^2}$

Differential Equations (MTH401)

Question: 36 (Marks: 1)

The differential operator form of the differential equation $|a|y'' + by' - 2y = e^{|x|}$ for $a < 0$ is _____.

Choices:

☒ $(-aD^2 + bD - 2)y = e^{|x|}$

☐ $(aD^2 + bD - 2)y = e^x$

☐ $(-aD^2 + bD - 2)y = e^{-x}$

☐ $(aD^2 + bD - 2)y = e^{|x|}$

Differential Equations (MTH401)

Question: 37 (Marks: 1)

To reduce Cauchy-Euler differential equation into a differential equation with constant coefficients we often utilize the substitution

Choices:

☒ $y = x^3$

☐ $y = e^{-t}$

☐ $y = e^t$

☐ None of these

Solution of the differential equation $4y'' + y = 0$, is

Choices:

☒ $y(x) = c_1 \cos\left(\frac{x}{2}\right)$

☐ $y(x) = c_1 \sin\left(\frac{x}{2}\right)$

☐ None of them

Question: 39 (Marks: 1)

The Differential Equation $10x^2y'' + 2xy' - y = 0$ has singularity at

Choices:

☐ $x = \pm i$

☒ $x = 0$

☐ $x = \pm 1$

☐ $x = 2$

Differential Equations (MTH401)

Question: 40 (Marks: 1)

The Differential Equation $(x^2 - 25)y'' + 30xy' - 2y = 0$ has singularity at

Choices:

☐ $x = \pm 5$

☒ $x = \pm 4$

☐ $x = \pm 3$

☐ $x = \pm 2$

Differential Equations (MTH401)

Question: 50 (Marks: 5)

Find the general solution of the given differential equation on $(0, \infty)$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \frac{1}{25}\right)y = 0$$

Answer:



Differential Equations (MTH401)

Question: 49 (Marks: 5)

An 8 lb weight stretches a spring 4 ft. Assuming that a damping force numerically equals to three times the instantaneous velocity acts on the system. If the weight is released from the equilibrium position with an upward velocity of 3 ft / sec. Determine the equation of the differential equation of motion of the free damped motion.

Answer:



Differential Equations (MTH401)

Question: 48 (Marks: 5)

Show that vectors $x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-4t}$ and $x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t}$ are linearly dependent.

Answer:



Differential Equations (MTH401)

Question: 46 (Marks: 3)

Find the Wronskian of the solutions of a system

$$X_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t} \text{ and } X_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t}$$

Answer:



Differential Equations (MTH401)

Question: 47 (Marks: 5)

State the procedure to solve a differential equation by using Frobenius Theorem.

Answer:



Differential Equations (MTH401)

Question: 44 (Marks: 3)

Write the solution of the given differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 11x = 0.$$

Answer:



If $m_1 = 4 + i$ and $m_2 = 4 - i$ are the roots of Cauchy Euler differential equation, then find the general solution.

Answer:



Differential Equations (MTH401)

Question: 43 (Marks: 3)

Write the homogenous system of differential equations

$$2x' - 5x + y' = 5e^t$$

$$x' - x + y' = e^t$$

In decoupled form.

Answer:



Question: 42 (Marks: 3)

Find the Eigen values of the given matrix

$$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}$$

Answer:



Differential Equations (MTH401)

Question: 41 (Marks: 3)

When roots of indicial equation differ by a positive integer then explain the case when $r_1 = r_2$, where r_1 and r_2 are roots of the indicial equation.

Answer:



Question 49

An 8 lb weight stretches a spring 4 ft. Assuming that a damping force numerically equals to three times the instantaneous velocity acts on the system. If the weight is released from the equilibrium position with an upward velocity of 3 ft/sec. Determine the equation of the differential equation of motion of the free damped motion

✓ Answer:

The differential equation of motion is given by:

$$my'' + by' + ky = 0$$

where m = mass, b = damping coefficient, k = spring constant

Given:

$m = 1/32$ (1 lb = 1/32 slugs), $b = 3$, $k = 4/4 = 1$ (since 4 ft is stretched by 1 lb)

So, the differential equation becomes:

$$(1/32)y'' + 3y' + y = 0$$

Question 47

State the procedure to solve a differential equation by using Frobenius Theorem

Answer:

The Frobenius Theorem is a method for solving differential equations of the form:

$$x^2 y'' + x p(x) y' + q(x) y = 0$$

where $p(x)$ and $q(x)$ are analytic functions.

Procedure:

1. Assume a solution of the form: $y = x^r \sum [a_n x^n] \text{ (n=0 to } \infty \text{)}$
2. Substitute this assumed solution into the differential equation.
3. Equate the coefficients of like powers of x to obtain a recurrence relation for the coefficients a_n .
4. Solve the recurrence relation to obtain the coefficients a_n .
5. Use the coefficients a_n to obtain the solution y .

Question 50

Find the general solution of the given differential equation on $(0, \infty)$ $x^2 d^2y/dx^2 + xdy/dx + (x^2 - 1/25)y = 0$

Answer:

This is a Bessel's equation of order $1/5$.

The general solution is given by:

$$y = c_1 J_{1/5}(x) + c_2 Y_{1/5}(x)$$

where $J_{1/5}(x)$ and $Y_{1/5}(x)$ are Bessel functions of order $1/5$, and c_1 and c_2 are arbitrary constants.

Question 41

When roots of indicial equation differ by a positive integer then explain the case when $r_1=r_2$, where r_1 , and r_2 are roots of the indicial equation.

Answer:

When $r_1 = r_2$, it means that the roots are repeated. In this case, the general solution is given by:

$$y = c_1 x^{r_1} + c_2 x^{r_1} \ln(x)$$

where c_1 and c_2 are arbitrary constants.

Question 42

Find the Eigen values of the given matrix

$$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}$$

Answer:

The characteristic equation is given by:

$$|A - \lambda I| = 0$$

where I is the identity matrix.

Expanding the determinant, we get:

$$(1-\lambda)(-7-\lambda) + 16 = 0$$

Simplifying, we get:

$$\lambda^2 + 6\lambda + 3 = 0$$

Solving for λ , we get:

$$\lambda = \frac{-6 \pm \sqrt{36-12}}{2}$$

$$\lambda = \frac{-6 \pm \sqrt{24}}{2}$$

$$\lambda = \frac{-6 \pm 2\sqrt{6}}{2}$$

$$\lambda = -3 \pm \sqrt{6}$$

Therefore, the Eigen values are $-3 + \sqrt{6}$ and $-3 - \sqrt{6}$

Question 44

Write the solution of the given differential equation

$$d^2x/dt^2 + 6dx/dt + 11x = 0.$$

Answer:

The characteristic equation is given by:

$$m^2 + 6m + 11 = 0$$

Solving for m , we get:

$$m = (-6 \pm \sqrt{36-44})/2$$

$$m = (-6 \pm \sqrt{-8})/2$$

$$m = (-6 \pm 2i\sqrt{2})/2$$

$$m = -3 \pm i\sqrt{2}$$

Therefore, the general solution is given by:

$$x(t) = e^{(-3t)} * (c_1 \cos(\sqrt{2}t) + c_2 \sin(\sqrt{2}t))$$

where c_1 and c_2 are arbitrary constants.

Question 45

If $m_1 = 4 + i$ and $m_2 = 4 - i$ are the roots of Cauchy Euler differential equation, then find the general solution.

Answer:

The general solution is given by:

$$y = c_1 x^{(4+i)} + c_2 x^{(4-i)}$$

where c_1 and c_2 are arbitrary constants.

Using Euler's formula, we can rewrite the solution as:

$$y = x^4 * (c_1 e^{(i \ln(x))} + c_2 e^{(-i \ln(x))})$$

Simplifying, we get:

$$y = x^4 * (c_1 \cos(\ln(x)) + c_2 \sin(\ln(x)))$$

where c_1 and c_2 are arbitrary constants.

MTh401 Final Term key 2025



01	C	11	B	21	C	31	C
02	B	12	B	22	C	32	A
03	A	13	D	23	B	33	C
04	A	14	A	24	---	34	A
05	D	15	C	25	---	35	--
06	B	16	B	26	B	36	A
07	A	17	B	27	C	37	D
08	B	18	B	28	A	38	D
09	A	19	B	29	A	39	B
10	B	20	D	30	B	40	A

